

Lab 11

Router on a Stick

Inter-VLAN Routing | CCNA 200-301

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ROAS Method

3 VLANs

Packet Tracer

CCNA 200-301

LAB 11 — ROUTER ON A STICK (ROAS)

[PURPOSE] Router on a Stick (ROAS) allows one physical router interface to route traffic between multiple VLANs using sub-interfaces. Each sub-interface acts as the default gateway for its VLAN — enabling all VLANs to communicate through a single cable connected to the router.

Key Concept — Why ROAS?

Without ROAS	With ROAS
Need 1 router port per VLAN	Only 1 physical interface needed
Expensive — router ports limited	Cost effective solution
Not scalable	Easy to add more VLANs
Not used in modern networks	Common in small networks

Topology

```
[ Router R1 ]
Fa0/0.10 → 192.168.10.1 (VLAN 10 Gateway)
Fa0/0.20 → 192.168.20.1 (VLAN 20 Gateway)
Fa0/0.30 → 192.168.30.1 (VLAN 30 Gateway)
      |
      Fa0/0 (Trunk)
      |
    [ Switch S1 ]
    /   |   \
Fa0/1 Fa0/2 Fa0/3
  |    |    |
PC1  PC2  PC3
VLAN10 VLAN20 VLAN30
HR   Finance IT
```

IP Address Plan

Router R1	Fa0/0.10	192.168.10.1/24	VLAN 10
Router R1	Fa0/0.20	192.168.20.1/24	VLAN 20
Router R1	Fa0/0.30	192.168.30.1/24	VLAN 30
PC1 (HR)	NIC	192.168.10.11/24	VLAN 10
PC2 (Finance)	NIC	192.168.20.11/24	VLAN 20
PC3 (IT)	NIC	192.168.30.11/24	VLAN 30
Switch S1	VLAN 1	192.168.10.2/24	Mgmt

SWITCH S1 — CONFIGURATION

Step 1 — Create VLANs

```
S1> enable
S1# configure terminal
S1(config)# vlan 10
S1(config-vlan)# name HR
S1(config-vlan)# exit
S1(config)# vlan 20
S1(config-vlan)# name FINANCE
S1(config-vlan)# exit
S1(config)# vlan 30
S1(config-vlan)# name IT
S1(config-vlan)# exit
```

Step 2 — Assign Access Ports

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# exit
S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# exit
S1(config)# interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 30
S1(config-if)# exit
```

Step 3 — Configure Trunk Port to Router

```
S1(config)# interface fa0/24
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,30
S1(config-if)# exit
S1(config)# end
S1# write memory
```

WARNING: The port connecting Switch to Router MUST be trunk — otherwise VLANs cannot reach the router!

ROUTER R1 — CONFIGURATION

Step 1 — Enable Physical Interface

```
R1> enable
R1# configure terminal
R1(config)# interface fa0/0
R1(config-if)# no shutdown
R1(config-if)# exit
```

TIP: Physical interface needs no shutdown — but NO ip address on physical interface!

Step 2 — Create Sub-interface for VLAN 10

```
R1(config)# interface fa0/0.10
R1(config-subif)# encapsulation dot1q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# exit
```

WARNING: encapsulation dot1q 10 is MANDATORY — without it sub-interface will not work!

Step 3 — Create Sub-interface for VLAN 20

```
R1(config)# interface fa0/0.20
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# exit
```

Step 4 — Create Sub-interface for VLAN 30

```
R1(config)# interface fa0/0.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# ip address 192.168.30.1 255.255.255.0
R1(config-subif)# exit
R1(config)# end
R1# write memory
```

PC CONFIGURATION

PC1 HR	192.168.10.11	255.255.255.0	192.168.10.1	10
PC2 Finance	192.168.20.11	255.255.255.0	192.168.20.1	20
PC3 IT	192.168.30.11	255.255.255.0	192.168.30.1	30

WARNING: PC Default Gateway = Router sub-interface IP — NOT switch IP!

VERIFICATION & TESTING

On Router — Check Interfaces:

```
R1# show ip interface brief
```

Interface	IP-Address	Status	Protocol
FastEthernet0/0	unassigned	up	up
Fa0/0.10	192.168.10.1	up	up OK
Fa0/0.20	192.168.20.1	up	up OK
Fa0/0.30	192.168.30.1	up	up OK

On Router — Check Routing Table:

```
R1# show ip route
```

```
C 192.168.10.0/24 is directly connected, Fa0/0.10
C 192.168.20.0/24 is directly connected, Fa0/0.20
C 192.168.30.0/24 is directly connected, Fa0/0.30
```

On Switch — Check Trunk:

```
S1# show interfaces trunk
```

Port	Mode	Encapsulation	Status
Fa0/24	on	802.1q	trunking OK

Port	Vlans allowed on trunk
Fa0/24	10,20,30

Ping Test — Different VLANs:

```
PC1 (VLAN10) ping PC2 (VLAN20):
PC> ping 192.168.20.11
Reply from 192.168.20.11    Inter-VLAN Working!

PC1 (VLAN10) ping PC3 (VLAN30):
PC> ping 192.168.30.11
Reply from 192.168.30.11    Inter-VLAN Working!
```

KEY POINTS & EXAM TIPS

ROAS Full Form	Router on a Stick
Sub-interface	Fa0/0.10, Fa0/0.20, Fa0/0.30
Encapsulation	dot1q [vlan number] — Mandatory!
Physical Interface	No IP — just no shutdown

Trunk Port	Switch to Router must be trunk
PC Gateway	Router sub-interface IP
Verify	show ip interface brief

CCNA Exam Tips:

```
Sub-interface number should match VLAN number (best practice)
encapsulation dot1q command is MANDATORY
Physical interface needs no shutdown – NO ip address!
Switch port to router MUST be trunk
PC default gateway = Router sub-interface IP – not switch!
show ip route – C means directly connected network
```

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